IGCSE 0607 Practice Questions — E1.3

Powers and roots

Total Questions: 6 — **Total Marks:** 34 Marks

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1 **(a)** Work out the value of

$$\sqrt{169} + 2^3$$

..... **[2]**

(b) Work out the value of

$$4^2 \times \sqrt[3]{125}$$

..... **[2]**

2 Solve the following equations.

(a)

$$\sqrt[3]{3x+4} = 4$$

$x =$ ..... **[3]**

(b)

$$x^2 = 4^3 + 6^2$$

$x =$ ..... , or $x =$ ..... **[2]**

(c)

$$\sqrt{x} + 2\sqrt{x} = 21$$

$x =$ ..... **[2]**

3 **(a)** Calculate the value of

$$\frac{\sqrt[3]{450} + 12.5^2}{3.4^3 - \sqrt{80}}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

(c) Find the largest integer n such that $n^4 < 2000$.

$n =$ ..... **[2]**

4 **(a)** Write down the value of

$\sqrt{196}$

..... **[1]**

(b) Work out the value of

$4^3 - \sqrt[3]{125}$

..... **[2]**

(c) Calculate the exact value of

$\sqrt{13^2 - 12^2}$

..... **[2]**

5 **(a)** **(i)** Calculate the value of

$$\sqrt[3]{8.5^2 - 4.1^3}$$

Write down your full calculator display.

..... **[2]**

(ii) Write your answer to part (a)(i) correct to 3 significant figures.

..... **[1]**

(b) Calculate the exact value of

$$\frac{4.8^2}{\sqrt{0.0576}}$$

..... **[2]**

6 **(a)** Find the integer k such that

$$\sqrt[3]{k} = 4^2 - 3^2 - 2$$

$k =$ ..... **[2]**

(b) Calculate the exact value of

$$\frac{\sqrt{10^2 - 6^2}}{\sqrt[3]{125} - 3^2}$$

..... **[3]**

(c) Given that $y > 0$ and $\sqrt[3]{y} + \sqrt[3]{y} = 6$, work out the value of y^2.

..... **[3]**